MILITARY SPECIFICATION SHEET

ELECTRON TUBE, MAGNETRON

TYPE 6410A

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Fixed nominal frequency of 2,805 MHz, 4.5 mW peak power output, with integral magnet

ABSOLUTE RATINGS:

Independent

Parameter:	If	epy	tk	VSWR	T(anode)	T(bushing)	Output pressurization
Unit:	A	kv	sec	---	°C	°C	psia
Maximum:	80 (Note 2)	75	---	1.5 (Note 3)	100 (Note 4)	200 (Note 4)	50 (Notes 5 and 6)
Minimum:	68	---	180	---	---	---	35

Dependent

Parameter:	ib	Pi	pi	Du	tpc
Unit:	a	kW	Mw	---	μs
Maximum:	135	10.0	10	0.001	2.2
Minimum:	90	---	---	---	1.8 (Note 7)

PHYSICAL CHARACTERISTICS:

Dimensions:	See figure 1	Cooling:	See notes 4 and 9
Mounting position:	See note 11	Cathode:	Unipotential (see note 8)
Mounting support:	By means of holes in magnet	Magnet isolation:	See note 10
Weight:	58 pounds, approximate		

TEST CONDITIONS: Note 6

Parameter:	If (preheat)	If (radiate)	Ib	VSWR	T (anode)	tk	tpc	tfc	tfv	trv	trc	Du
Unit:	A	A	mAdc	---	°C	sec	μs	μs	μs	μs	μs	---
Test condition 1:												
Maximum:	80	76	---	---	90	180	2.2	0.6	2.5	0.5	0.35	---
:	---	---	130	1.1	---	---	---	---	---	---	---	0.001
Minimum:	75 [Notes 2, 14, and 15]	70	---	---	50	---	---	← Note 18 →				---
Test condition 2:												
Maximum:	80	76	---	---	90	---	2.2	0.6	2.5	0.5	0.35	---
:	---	---	104	1.1	---	---	---	---	---	---	---	0.0008
Minimum:	75 [Notes 2, 14, and 15]	70	---	---	50	180	---	← Note 18 →				---

GENERAL:

Qualifications - Required

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METHOD	REQUIREMENT OR TEST	NOTES	TEST	CONDITIONS	SYMBOL	LIMITS		UNIT
						MINIMUM	MAXIMUM	
	Qualification inspection							
4027	Temperature coefficient	4	1, 2	T = +40°C to +70°C	ΔF/ΔT	---	-0.15	MHz/°C
1031	High-frequency vibration	---	---	No voltages	---	---	---	---
1042	Shock	---	---	15 G; 11 ms; three planes one direction	---	---	---	---
---	Fixed frequency	4	1		F	2,750	2,860	MHz
4306	Pulse voltage	---	---		epy	65	75	kv
4250	Power output	---	---		Po	4,500	---	W
4308	RF bandwidth	19	---		BW	---	1.5	MHz
4315	Pulse stability	16	1		MP	---	0.5	%
4311	Frequency pushing figure	17	1, 2		ΔF/Δib	---	0.025	MHz/A
	Quality conformance inspection, part 1	1	---					
4003	Pressurizing	---	---	50 lb/sq in. absolute (max), there shall be no leaks at the output	---	---	---	---
1261	Filament voltage	---	---	If = 76 A; tk = 600 sec (min)	Ef	6.5	8.8	V
---	Fixed frequency	4	2		F	2,750	2,860	MHz
4306	Pulse voltage	---	---		epy	65	75	kv
4250	Power output	---	---		Po	3,600	---	W
4308	RF bandwidth	19	---		BW	---	1.5	MHz
4315	Pulse stability	16	2		MP	---	0.5	%
	Quality conformance inspection, part 2							
1031	Low-frequency vibration	---	---	No voltages	---	---	---	---
4310	Frequency pulling figure	---	2		ΔF	---	15	MHz
	Quality conformance inspection, part 3							
---	Life test	12, 13	2	Group D	t	400	---	hrs

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METHOD	REQUIREMENT OR TEST	NOTES	TEST	CONDITIONS	SYMBOL	LIMITS		UNIT
						MINIMUM	MAXIMUM	
	Quality conformance inspection, part 3 -Continued							
---	Life-test end points:	---	---		---	---	---	---
---	Fixed frequency	4	2		F	2,750	2,860	MHz
4250	Power output	---	1		Po	3,600	---	W
			2		Po	2,880	---	W
4308	RF bandwidth	19	---		BW	---	2.0	MHz
4315	Pulse stability	16	2		MP	---	1.0	%

NOTES:

1. Unless otherwise specified, the AQL for quality conformance inspection, part 1, shall be 1.0 percent, inspection level II.
2. The maximum value specified is for a nonoscillating condition. Heater surge current should not be allowed to exceed 100 amperes.
3. Frequency skipping or unstable operation may be encountered at some phase positions when the mismatch occurs at the end of a "long" line.
4. The anode and bushing temperature shall be measured at the points indicated on figure 1. The temperature shall be measured after thermal equilibrium has been reached.
5. During operation, the gas used in pressurization shall provide insulating properties at least equal to that of clean dry air at the pressures indicated.
6. Under test condition 1, or any other operating conditions in excess of 0.0008 duty cycle, it is essential to provide a stream of air (5 cfm minimum) to cool the magnetron output window in order to prevent damage due to thermal stresses.
7. The characteristics of the applied pulse shall be those which result in proper starting and oscillation. The rate of pulse voltage rise, the percentage of pulse voltage ripple, and the rate of pulse voltage fall are among the more important considerations. Negative post-pulse voltage oscillations may cause leakage current and noise.
8. The cathode terminal and high-voltage bushing shall be insulated by immersion in oil having the equivalent electrical characteristics of Federal Specification VV-I-530.
9. At all times the anode temperature shall be maintained at least 5°C above the freezing point of the anode liquid coolant. Coolant pressure shall not exceed 55 pounds per square inch absolute, but sufficient pressure of coolant shall be maintained to provide cooling of the anode to the temperature limits specified.
10. In handling and mounting the magnetron, care shall be exercised to prevent demagnetization. Ferromagnetic materials or energized magnets shall be kept more than 12 inches from the tube. Maintain 18 inches spacing (center-to-center) to any other 6410A tube. Do not place an unpacked 6410A tube on a case containing another 6410A tube.
11. The tube shall be mounted with the cathode vertical within 15° and with the end of the cathode bushing pointing in the downward direction.
12. Power input to the tube during life test shall be cycled according to the following schedule:

    Preheat (heater) - 5 minutes
    Oscillate (RF) - 4 hours
    All voltages off - 1 hour

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NOTES: - Continued

13. The interpretation of the VSWR shall be as specified in method 4310. The standing wave introducer shall be cycled continuously through a one-half wavelength line at a minimum rate of one cycle/hour.

14. The heater current should be reduced at 75 percent of the maximum power input. This reduced heater current shall be maintained at ±3.5 percent of the specified value.

15. Each tube shall be labeled to indicate If for both preheat and radiate conditions.

16. The interpretation of the VSWR shall be as specified in method 4310. The missing pulse rate shall be counted during the last 5 minutes of a test interval not to exceed 15 minutes.

17. The peak anode current shall be varied at more than 50 Hz through a range exceeding 20 percent, but not exceeding 45 percent of the peak value obtained under test condition 1 or test condition 2 conditions; the average current being held at test condition 1 or test condition 2 value. This test may be performed under either test condition 1 or test condition 2 conditions. The frequency shift per ampere shall be within the limit specified.

18. This test shall be conducted for only one set of conditions within the limits specified for the test condition specified. The pulse characteristics shall be measured as follows:

    tfc and tfv - 0 to 85 percent point
    trc and trv - 20 to 85 percent point

    No spike ripple shall exceed ±7 percent of the average peak value of the voltage pulse. Inverse voltage shall not exceed 35 percent of the forward voltage. There shall be no negative post-pulse voltage.

19. The rf bandwidth shall be within the limits specified when a VSWR of 1.5:1 is introduced into the load, the phase being adjusted for maximum spectrum width.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army - EL, MU
Air Force - 11, 17, 80
DSA - ES

User activities:
Army - WC
Navy - AS, OS, MC, CG, SH

Preparing activity: Army - EL

Agent: DSA - ES

(Project 5960-2425-61)

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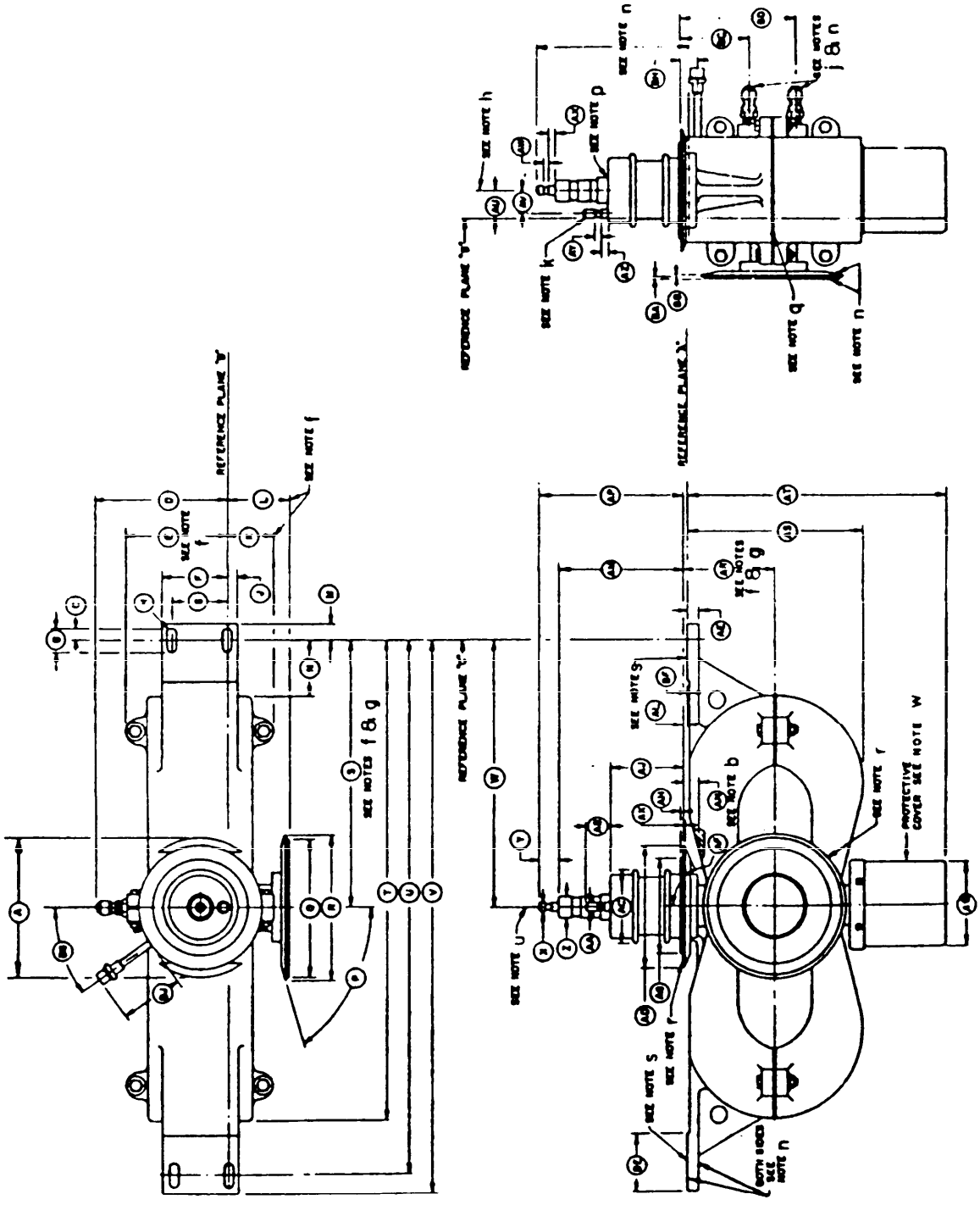


FIGURE 1. Outline drawing of electron tube type 6410A.

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
	Quality conformance inspection, part 1 (see note 1)	
E	.860 (21.84)	.890 (22.61)
J	.425 (10.80)	.450 (11.43)
P		2.437 (61.90)
S	.190rad(4.83)	.210rad(5.33)
U		.437 (11.10)
V	2.195 (55.75)	2.445 (62.10)
W		.750 (19.05)
Z	5.025dia(127.64)	5.051dia(128.30)
AB	5.285dia(134.24)	5.315dia(135.00)
AE	9.662 (245.41)	9.962 (253.03)
AF	19.437 (493.70)	19.813 (503.25)
AG		20.625 (523.88)
AK		.344dia(8.74)
AU		.312dia(7.92)
AV		2.875dia(73.03)
AW	4.485dia(113.92)	4.515dia(114.68)

Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
	Quality conformance inspection, part 1 (see note 1) - Continued	
AJ		3.500 (88.90)
AL		.300 (7.62)
AM		5.000(127.00)
AN	.385 (9.78)	
AP		6.000(152.40)
AQ		3.250dia(82.55)
AR	3.250 (82.55)	3.500 (88.90)
AS		6.500(165.10)
AT		9.750(247.65)
	Quality conformance inspection, part 2	
A	4.812dia(122.22)	
E		3.875 (98.43)
G	1.985 (50.42	2.015 (51.18)
K		1.812 (46.02)
N	1.250 (31.75)	
T		18.500(469.90)
BE	2.000 (50.80)	
BF		.062 (1.57)

Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
	Reference dimensions	
D	4.917(124.89)	
P	70°	
W	9.812(249.22)	
Y	.674 (17.12)	
Z	.740 dia (18.80)	
AB	.875 (22.23)	
AE	.375 (9.53)	
AF	70°	
AG	3.485 dia (88.52)	
AH	.125 (3.18)	
AK	.063 (1.60)	
AU	1.000 (25.40)	
AV	.850 (21.59)	
AW	.187 (4.75)	
AX	.250 (6.35)	
AY	.187 (4.75)	
AZ	.250 (6.35)	
BA	.094 (2.39)	
BB	.063 (1.60)	
BC	2.620 (66.55)	
BD	4.120(104.65)	
BG	35°	
BH	.685 (17.40)	
BJ	2.250 (57.15)	

NOTES:
a. Dimensions without tolerances are for information and are not required for inspection purposes.
b. Applies to dimension A.
c. Reference plane A is defined as that plane which passes along face of mounting brackets.
d. Reference plane B is defined as that plane which passes through centers of mounting slots as shown, and which is perpendicular to plane A.
e. Reference plane C is defined as that plane which passes through centers of mounting slots as shown, and which is mutually perpendicular to plans A and B.
f. Includes angular and lateral deviations.
g. Refers to center line of waveguide.
h. Parts located on this line may vary from true location by 0.125.
j. Coupling plug, Hansen B3-T20, or equivalent.
k. Common cathode connection.
m. All soldered joints on output section shall provide hermetic seal.
n. The following surfaces shall be free from paint: Front and rear faces of oil sealing sleeve, parts above flange, front and rear faces of output flange, exhaust cap, and mounting surfaces of magnets and coolant plug connectors.
p. Cathode stem temperature measured at this point.
q. Anode temperature shall be measured at right side (facing output) of output transformer by inserting thermocouple plug as shown on figure 2.
r. Equipment flanges shall mate with oil sealing flange and output flange by means of flexible union. Oil sealing flange and output flange shall not be used to support tube. Entire weight of tube shall be supported by magnet mounting brackets.
s. These surfaces shall be coplanar within 0.032. This feature shall be inspected on a qualification basis.
t. Paint with heat resistant, noncorrosive paint.
u. Parts located on this line may vary from true location by 0.188.
v. Magnet may be of two-piece or four-piece construction.
w. Protective shipping covers shall be provided.

FIGURE 1. Outline drawing of electron tube type 6410A - Continued.

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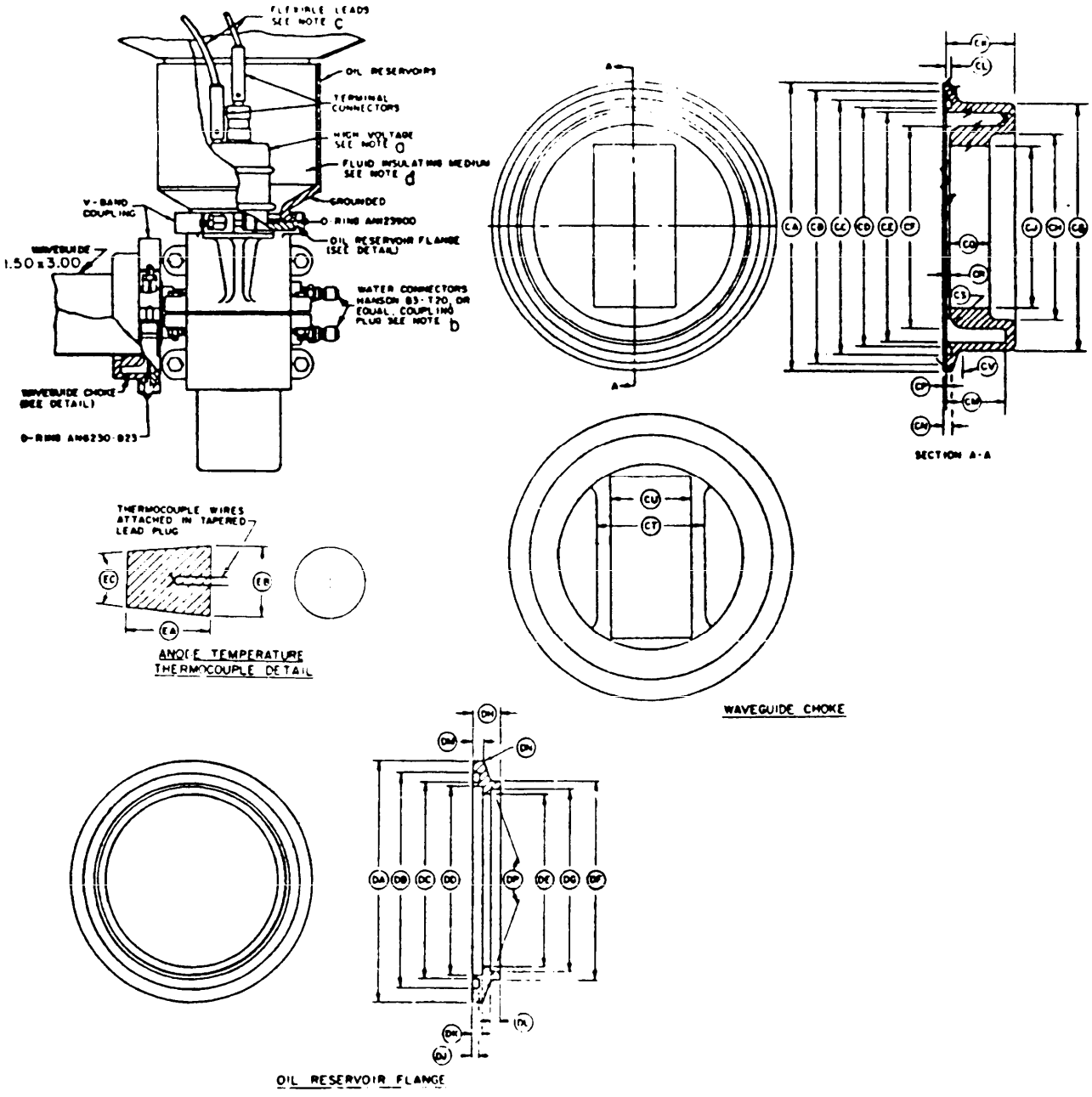


FIGURE 2. Installation details.

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
CB	5.052dia (128.32)	5.062dia (128.57)
CC	4.684dia (118.97)	4.688dia (119.08)
CD	4.382dia (111.30)	4.386dia (111.40)
CE	4.260dia (118.20)	4.270dia (118.46)
CF	3.760dia (95.50)	3.770dia (95.76)
CJ	3.000 (76.20)	3.010 (76.45)
CL	.113 (2.87)	.118 (3.00)
CM	1.115 (28.32)	1.125 (28.58)
CN	.169 (4.29)	.173 (4.39)
CP	.065 (1.65)	.070 (1.78)
CR	.045 (1.14)	.055 (1.40)
CU	1.495 (37.97)	1.505 (38.23)
CV	19° 30'	20° 30'
DA	4.490dia (114.05)	4.510dia (114.55)

Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
DB	4.000dia (101.60)	4.010dia (101.85)
DC	3.625dia (92.08)	3.635dia (92.33)
DD	3.500dia (88.90)	3.510dia (89.15)
DE	3.195dia (81.15)	3.205dia (81.41)
DF	3.675dia (93.35)	3.685dia (93.60)
DG	3.395dia (86.23)	3.405dia (86.49)
DH	.528 (13.41)	.532 (13.51)
DJ	.119 (3.02)	.123 (3.12)
DK	.156 (3.96)	.160 (4.06)
DL	.183 (4.65)	.189 (4.80)
DM	.195 (4.95)	.199 (5.05)
DP	139° 30'	140° 30'
EA	.160 (4.06)	.170 (4.32)
EB	.130dia (3.30)	.145dia (3.68)
EC	10° 20'	11° 0'

Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Reference dimensions		
CA	5.312 dia (134.92)	
CG	4.562 dia (115.87)	
CH	3.438 dia (87.33)	
CK	1.281 (32.54)	
CQ	.750 (19.05)	
CS	.250 rad (6.35)	
CT	2.000 (50.80)	
DN	.031 rad (.79)	

NOTES:
a. Minimum distance from high voltage to ground shall be 1.500 through oil and 10.000 through air, at atmospheric pressure.
b. Mates with socket, Hanson 3-S20, or equivalent.
c. Leads shall conduct 100 RMS amperes minimum.
d. Fluid insulating medium shall be Univolt 35 transformer oil, or equivalent.
e. All corners .125 rad., except as shown.

FIGURE 2. Installation details - Continued.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-E-1/665D ELECTRON TUBE, MAGNETRON

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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